CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of M. Pharm (Pharmaceutical Technology)

University Theory Examination January 2018

PHPT002 PHARMACEUTICAL PRODUCT DEVELOPMENT

Date: 09.01.2018, Tuesday **Time:10:00 a.m. to 1:00 p.m.** **Maximum Marks: 70**

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

Q.1	**A)**	Explain the concept of preformulation studies.	5
	B)	Describe various preformulation parameters for biotechnological products.	5
Q.2	**A)**	Write a note on: Standardization of excipients.	5
	B)	Describe the importance of *in-vitro* dissolution studies for pharmaceutical products.	5
		OR	
Q.2	**A)**	Explain in brief: Phase solubility and Inclusion complex.	5
	B)	Write a detailed note on BCS Classification and describe its significance in dissolution process.	5
Q.3		**Answer the following (Any three)**	**15**
	(A)	Explain: Thermodynamics of diffusion.	
	(B)	Explain with example the importance of similarity and dissimilarity factors in dissolution study of formulations.	
	(C)	Describe Fick's first law of diffusion.	
	(D)	Enumerate and explain the various methods for the characterization of polymorphs and solvates.	

SECTION-II

Q.4	(i) What is the importance of stability indicating assay in stability evaluation of the product?	**05**
	(ii) Describe the mechanism and importance of 'Accelerated stability study' in detail in context to the stability of the products.	**05**
Q.5	Describe 'Cyclodextrins' as pharmaceutical excipients with various examples and applications.	**10**
	OR	
Q.5	Classify Polymers. Explain the importance of polymers in Novel drug delivery system with examples. Explain that how bio-degradable polymers can be characterized and evaluated?	**10**

Q.6 **Answer the following (Any three)** 15

(A) Write a brief note on: Medical Prosthetics.

(B) Explain the following indicating their applications:

(i) Co-processed excipients (ii) Ion-exchange resins

(C) Which pharmacopoeial dissolution apparatus are used for the dissolution study of modified release dosage forms? Explain in detail its construction and working.

(D) Write a brief note on: ICH guidelines for quality of product.
